Introduction to the Mathematics of the XY-Spin Chain

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Abstract

In the following we present an introduction to the mathematical theory of the XY spin chain. The importance of this model lies in the fact, first understood by Lieb, Schultz and Mattis in [5], that the XY spin chain is one of very few "exactly solvable" models in the theory of quantum many-body systems. Lieb, Schultz and Mattis considered the constant coefficient case. In the variable coefficient case considered here, "exactly solvable" should be understood as "reducible to an effective one-particle Hamiltonian". The key method behind this is the Jordan-Wigner transform, which allows to map the XY chain to a free Fermion system.

1 A warm-up: The isotropic XY-spin chain

For a positive integer n, consider the Hamiltonian

$$H = -\sum_{j=1}^{n-1} \mu_j (\sigma_j^X \sigma_{j+1}^X + \sigma_j^Y \sigma_{j+1}^Y) - \sum_{j=1}^{n} \nu_j \sigma_j^Z, \tag{1}$$

acting in $\mathcal{H} = \bigotimes_{j=1}^n \mathbb{C}^2$. The first sum represents a chain of n spins with next neighbors interacting through the X and Y-Pauli matrices. The second sum models the effect of an additional transverse magnetic field acting on the spins.

The parameters μ_j and ν_j are real. We also assume $\mu_j \neq 0$, as otherwise the chain breaks into separate pieces which could be analyzed individually. In most of the physics literature on the XY chain these parameters are chosen as constants $\mu \neq 0$ and ν, but we want to stress here that the methods used also work in the variable coefficient case. The word "isotropic" refers to the fact that the two interaction terms, in the X and Y-Pauli matrices, respectively, have equal weight. We start with this case, mostly because it is easier. The anisotropic XY chain will be considered in Section 5 below.

To be specific, let

$$\sigma_j^\bullet = I \otimes \ldots \otimes I \otimes \sigma^\bullet \otimes I \otimes \ldots \otimes I, \quad \bullet \in \{X, Y, Z\}, \tag{2}$$

with non-trivial entries in the j-th component of the tensor product, and choose the standard representation

$$\sigma^X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3}$$

of the Pauli matrices. We note that, using $\{A, B\} = AB + BA$ to denote anti-commutators,

$$(\sigma^X)^2 = (\sigma^Y)^2 = (\sigma^Z)^2 = I, \quad \{\sigma^X, \sigma^Y\} = \{\sigma^X, \sigma^Z\} = \{\sigma^Y, \sigma^Z\} = 0. \tag{4}$$

We will frequently work with the lowering and raising operators

$$a := \frac{1}{2}(\sigma^X - i\sigma^Y) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \tag{5}$$

and

$$a^* = \frac{1}{2}(\sigma^X + i\sigma^Y) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{6}$$

Note that

$$a^*a = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \text{and} \quad aa^* = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{7}$$

are orthogonal projections. As above we write

$$a_j = I \otimes \ldots \otimes I \otimes a \otimes I \otimes \ldots \otimes I, \tag{8}$$

$$a_j^* = I \otimes \ldots \otimes I \otimes a^* \otimes I \otimes \ldots \otimes I. \tag{9}$$

This allows to express the Pauli matrices through

$$\sigma_j^X = a_j + a_j^*, \quad \sigma_j^Y = i(a_j - a_j^*), \quad \sigma_j^Z = 2a_j^*a_j - I. \tag{10}$$

The raising and lowering operators satisfy the mixed commutator and anti-commutator relations

$$\{a_j, a_j^*\} = I, \quad (a_j)^2 = (a_j^*)^2 = 0, \tag{11}$$

$$[a_j, a_k^*] = [a_j^*, a_k^*] = [a_j, a_k] = 0 \quad \text{for } j \neq k. \tag{12}$$

Using these identities one verifies

$$\sigma_j^X \sigma_{j+1}^X + \sigma_j^Y \sigma_{j+1}^Y = 2(a_j^* a_{j+1} + a_{j+1}^* a_j), \quad j = 1, \ldots, n-1, \tag{13}$$

and therefore H can be expressed as

$$H = -2\sum_{j=1}^{n-1} \mu_j (a_j^* a_{j+1} + a_{j+1}^* a_j) - \sum_{j=1}^{n} \nu_j (2a_j^* a_j - I). \tag{14}$$

2 The Jordan-Wigner transform

Lieb, Schultz and Mattis [5] were the first to realize that the Hamiltonian of the xy-spin chain is equivalent to an effective one-particle system, which, in cases where the Jacobi matrix M representing the one-particle system can be diagonalized (in particular for the constant coefficient case), makes the xy-spin chain explicitly solvable.

The first step in this diagonalization is a Jordan-Wigner transform, where a_j and a_j^* are replaced by a new set of so-called annihilation and creation operators (the meaning of which will become clearer in Section 3)

$$c_j := \sigma_1^Z \dots \sigma_{j-1}^Z a_j, \quad j = 1, \dots, n, \tag{15}$$

and

$$c_j^* = \sigma_1^Z \dots \sigma_{j-1}^Z a_j^*. \tag{16}$$

From this it follows readily that

$$c_j^2 = a_j^2 = 0, \quad (c_j^*)^2 = (a_j^*)^2 = 0, \quad c_j^* c_j = a_j^* a_j, \quad c_j c_j^* = a_j a_j^*, \tag{17}$$

and therefore $\{c_j, c_j^*\} = I$. In fact, the c_j and c_j^* satisfy the *canonical anti-commutation relations* (CAR),

$$\{c_j, c_k^*\} = \delta_{jk} I, \quad \{c_j, c_k\} = \{c_j^*, c_k^*\} = 0 \quad \text{for all } j, k = 1, \dots, n, \tag{18}$$

To verify this for $k \neq j$, use that $\{a_j, \sigma_j^Z\} = 0$.

To express H in terms of the operators c_j and c_k^*, note that

$$a_j^* a_{j+1} = c_j^* \sigma_j^Z c_{j+1} = c_j^*(2c_j^* c_j - I) c_{j+1} = -c_j^* c_{j+1} \tag{19}$$

and, taking adjoints,

$$a_{j+1}^* a_j = -c_{j+1}^* c_j. \tag{20}$$

This leads to

$$\begin{aligned} H &= 2\sum_{j=1}^{n-1} \mu_j (c_j^* c_{j+1} + c_{j+1}^* c_j) - 2\sum_{j=1}^{n} \nu_j c_j^* c_j + \sum_{j=1}^{n} \nu_j I \\ &= 2c^* M c + E_0 I. \end{aligned} \tag{21}$$

Here $c := (c_1, \dots, c_n)^t$, $c^* = (c_1^*, \dots, c_n^*)$, $E_0 := \sum_j \nu_j$ and M is the symmetric Jacobi matrix

$$M = (M_{jk})_{j,k=1}^n := \begin{pmatrix} -\nu_1 & \mu_1 & & \\ \mu_1 & \ddots & \ddots & \\ & \ddots & \ddots & \mu_{n-1} \\ & & \mu_{n-1} & -\nu_n \end{pmatrix}. \tag{22}$$

In the theory of the XY chain it turns out that M plays the role of an *effective one-particle Hamiltonian.* In principle, the many body Hamiltonian H (acting on a 2^n-dimensional Hilbert space) can be fully understood from properties of the one-particle Hamiltonian M (which acts on an n-dimensional Hilbert space).

3 Diagonalization of H

There is an orthogonal matrix U (with real entries) such that

$$UMU^t = \Lambda = \mathrm{diag}(\lambda_j), \tag{23}$$

and therefore

$$e^{-iMt} = U^t e^{-i\Lambda t} U = U^t \mathrm{diag}(e^{-i\lambda_j t}) U. \tag{24}$$

Define $b = (b_1, \ldots, b_n)^t$ by

$$b = Uc. \tag{25}$$

By the bi-linearity of $\{\cdot,\cdot\}$ and orthogonality of U it follows readily that b_j, $j = 1, \ldots, n$ satisfies the CAR. Moreover,

$$\begin{aligned} H &= 2c^* M c + E_0 I \\ &= 2c^* U^t \Lambda U c + E_0 I \\ &= 2b^* \Lambda b + E_0 I \\ &= \sum_{j=1}^{n} 2\lambda_j b_j^* b_j + E_0 I, \end{aligned} \tag{26}$$

i.e. in terms of the new creation and annihilation operators H takes the form of a Hamiltonian of a *free Fermion system*. Assuming knowledge of the λ_j, this operator can be explicitly diagonalized. This makes use of important algebraic properties of systems of operators satisfying the CAR. For a somewhat useful introduction or a reminder of these properties see [6]. For a brief treatment see Proposition II.6.2 in [8]. Some of them are:

- The operators $b_j^* b_j$, $j = 1, \ldots, n$, are pairwise commuting orthogonal projections, meaning that they can be simultaneously diagonalized.
- The intersection of the kernels of $b_j^* b_j$ (which is the same as the kernel of b_j) contains at least one non-trivial normalized vector Ω, i.e. $b_j^* b_j \Omega = 0$, $j = 1, \ldots, n$.
- For each $\alpha = (\alpha_1, \ldots, \alpha_n) \in \{0,1\}^n$ use successive *creation operators* b_j^* to define

  $$\psi_\alpha = (b^*)^\alpha \Omega := (b_1^*)^{\alpha_1} \ldots (b_n^*)^{\alpha_n} \Omega, \tag{27}$$

  which are an orthonormal system of common eigenvectors for the $b_j^* b_j$:

  $$b_j^* b_j \psi_\alpha = \begin{cases} 0, & \text{if } \alpha_j = 0, \\ \psi_\alpha, & \text{if } \alpha_j = 1. \end{cases} \tag{28}$$

- In the given case $\dim \mathcal{H} = 2^n$, and thus the vectors ψ_α form an orthonormal basis. In particular, the normalized vector Ω is unique up to a trivial phase. It is frequently referred to as the "vacuum vector".

The vectors ψ_α are eigenvectors for H,

$$H\psi_\alpha = \left(2\sum_{j:\alpha_j=1}\lambda_j + E_0\right)\psi_\alpha, \tag{29}$$

which gives the spectrum of H and, in particular, the ground state energy

$$2\sum_{j=1}^{n}\min\{0,\lambda_j\} + E_0 \tag{30}$$

and corresponding ground state $\varphi_0 = \psi_{\alpha^{(0)}}$, where

$$\alpha_j^{(0)} = \begin{cases} 1, & \text{if } \lambda_j < 0, \\ 0, & \text{else.} \end{cases} \tag{31}$$

which is non-degenerate if and only if $\lambda_j \neq 0$ for all j.

Note that $E_0 = \sum_j \nu_j = -\operatorname{tr} M = -\sum_j \lambda_j$. Using this we can re-write (29) and describe the spectrum of H as

$$\sigma(H) = \{2\sum_{j:\alpha_j=1}\lambda_j - \operatorname{tr} M : \alpha \in \{0,1\}^n\}. \tag{32}$$

We also note that the ψ_α are eigenvectors of the "total particle number" operator $\mathcal{N} := \sum_j b_j^* b_j$. In fact, by (28),

$$\mathcal{N}\psi_\alpha = N_\alpha\psi_\alpha, \quad N_\alpha := \#\{j : \alpha_j = 1\}. \tag{33}$$

The operator $\mathcal{N}$ could alternatively be defined by using the operators c_j or a_j since we have

$$\mathcal{N} = \sum_j b_j^* b_j = \sum_j c_j^* c_j = \sum_j a_j^* a_j, \tag{34}$$

where (25) and orthogonality of U is used. The latter allows for a more concrete description of the eigenspaces of $\mathcal{N}$: Writing

$$e_0 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{35}$$

for the canonical basis vectors of $\mathbb{C}^2$ and

$$e_\alpha := e_{\alpha_1} \otimes \ldots \otimes e_{\alpha_n}, \quad \alpha \in \{0,1\}^n \tag{36}$$

for the product basis of $\bigotimes^n \mathbb{C}^2$, it follows from $\mathcal{N} = \sum_j a_j^* a_j$ and (7) that $\mathcal{N} e_\alpha = N_\alpha e_\alpha$. Thus the eigenspace of $\mathcal{N}$ to any integer $k \in \{0, 1, \ldots, N\}$ is spanned by the set of all e_α with exactly k up-spins, and thus has dimension $\binom{n}{k}$.

The fact that H commutes with $\mathcal{N}$ implies that all these eigenspaces are invariant under H, which could also be verified directly from (14). Thus H conserves the number of up-spins in each basis state, which is frequently referred to as *conservation of particle number* and provides the reason for calling $\mathcal{N}$ the total particle number operator.

4 Dynamics of the creation and annihilation operators

For a bounded linear operator A on $\mathcal{H}$ we will write

$$\tau_t(A) := e^{itH} A e^{-itH} \tag{37}$$

for the Heisenberg dynamics under H.

We first observe that, for all $k = 1, \ldots, n$,

$$\tau_t(b_k) = e^{-2it\lambda_k} b_k \quad \tau_t(b_k^*) = e^{2it\lambda_k} b_k^*. \tag{38}$$

To see the first identity (the second follows by taking adjoints) check by Taylor expansion, using $b_k^2 = 0$, that

$$e^{2it\lambda_k b_k^* b_k} b_k = b_k, \tag{39}$$

and similarly, using $b_k b_k^* b_k = (-b_k^* b_k + I) b_k = b_k$,

$$b_k e^{-2it\lambda_k b_k^* b_k} = e^{-2it\lambda_k} b_k. \tag{40}$$

By combining this, we find (38) from

$$\begin{aligned} \tau_t(b_k) &= \prod_\ell e^{2it\lambda_\ell b_\ell^* b_\ell} b_k \prod_m e^{-2it\lambda_m b_m^* b_m} \\ &= e^{2it\lambda_k b_k^* b_k} b_k e^{-2it\lambda_k b_k^* b_k} \\ &= e^{-2it\lambda_k} b_k. \end{aligned} \tag{41}$$

Remark: A different proof of (38) uses that

$$\frac{d}{dt}\tau_t(b_k) = -i\tau_t([b_k, H]) \tag{42}$$

$$= -2i \sum_\ell \lambda_\ell \tau_t([b_k, b_\ell^* b_\ell]) \tag{43}$$

$$= -2i\lambda_k \tau_t([b_k, b_k^* b_k]) \tag{44}$$

$$= -2i\lambda_k \tau_t(b_k). \tag{45}$$

The unique solution of this differential equation with initial condition $\tau_0(b_k) = b_k$ is given by (38).

Using (25), (38) and (24), we find the dynamics of the c_k, working in convenient vector notation,

$$\begin{aligned} \tau_t(c) &= e^{itH} c e^{-itH} \\ &= e^{itH} U^t b e^{-itH} \\ &= U^t e^{itH} b e^{-itH} \\ &= U^t e^{-2it\Lambda} b \\ &= U^t e^{-2it\Lambda} U c \\ &= e^{-2iMt} c, \end{aligned} \tag{46}$$

or, in components,

$$\tau_t(c_j) = \sum_\ell v_{j\ell}(t) c_\ell, \tag{47}$$

where $v_{j\ell}(t) := \left(e^{-2iMt}\right)_{j\ell}$. By taking adjoints we get

$$\tau_t(c_j^*) = \sum_\ell \overline{v_{j\ell}(t)} c_\ell^*. \tag{48}$$

What this means is that information on the time evolution $v_{j\ell}(t)$ of the effective one-particle Hamiltonian M can be turned into information on the Heisenberg evolution $\tau_t(\cdot)$ under the many-body Hamiltonian H. However, a difficulty arises from the fact that the Jordan Wigner transform is non-local, meaning that the operators c_j act non-locally (i.e. on many spins simultaneously). In order to get information on the Heisenberg evolution of local operators such as a_j and a_j^* from this, one needs to find ways to "undo" the Jordan-Wigner transform, which is generally a non-trivial task.

One example where undoing of Jordan-Wigner turned out to be possible is described in [2]. There the case of a random magnetic field, described by choosing the parameters ν_j as i.i.d. random variables, is considered. In this case the effective Hamiltonian M becomes the Anderson model. Known results on dynamical localization for the latter allow to deduce a so-called zero-velocity Lieb-Robinson bound for the XY chain in random field, a result which is interpreted as absence of information transport in the spin chain [2].

5 The anisotropic xy-chain

We now consider an anisotropic generalization of (1), namely the Hamiltonian

$$H_\gamma = -\sum_{j=1}^{n-1} \mu_j[(1+\gamma_j)\sigma_j^X\sigma_{j+1}^X + (1-\gamma_j)\sigma_j^Y\sigma_{j+1}^Y] - \sum_{j=1}^{n} \nu_j\sigma_j^Z \tag{49}$$

in $\mathcal{H} = \bigotimes_{j=1}^n \mathbb{C}^2$. The additional parameters γ_j describe the anisotropy in the two interaction terms. We will generally assume $\gamma_j \in [0,1]$, where the choice $\gamma_j = 0$ recovers the isotropic XY chain and $\gamma_j = 1$ gives the quantum Ising chain, where only the interaction terms proportional to $\sigma_j^X\sigma_{j+1}^X$ appear. We use the notation H_γ, with γ being short for $(\gamma_j)_{j=1}^n$, to better distinguish the anisotropic XY chain from the isotropic XY chain considered above.

Below we provide a mathematically streamlined version of a transformation described in [5] for the constant coefficient case (and without magnetic field), adapted to the case of variable coefficients μ_j, γ_j and ν_j. In addition to the Jordan-Wigner transform this uses a "Bogoliubov transformation" (of which (25) can be considered as a special case) to reduce the Hamiltonian to a free Fermion system.

Lieb, Schultz and Mattis [5] considered the model (49) without magnetic field, constant $\mu_j = 1$ and constant $\gamma_j = \gamma \in (0,1)$. A constant transverse magnetic field was added in [3] and [1]. The Ising model case $\gamma = 1$ with constant transverse field was studied in [7].

Using facts from Section 1 as well as

$$\sigma_j^X\sigma_{j+1}^X - \sigma_j^Y\sigma_{j+1}^Y = 2(a_j a_{j+1} + a_{j+1}^* a_j^*) \tag{50}$$

we can express H_γ in terms of the operators a_j as

$$H_\gamma = -2\sum_{j=1}^{n-1}\mu_j[a_j a_{j+1}^* + a_{j+1}a_j^* + \gamma_j(a_j a_{j+1} + a_{j+1}^* a_j^*)] - \sum_{j=1}^{n}\nu_j(2a_j^* a_j - I). \tag{51}$$

With c_j as above, using (17), (19), (20) as well as

$$a_j a_{j+1} = c_j c_{j+1}, \quad a_{j+1}^* a_j^* = c_{j+1}^* c_j^*, \tag{52}$$

we get

$$\begin{aligned} H_\gamma &= -2\sum_{j=1}^{n-1}\mu_j[c_j c_{j+1}^* + c_{j+1}c_j^* + \gamma_j(c_j c_{j+1} + c_{j+1}^* c_j^*)] - \sum_{j=1}^{n}\nu_j(2c_j^* c_j - I) \\ &= -\sum_{j=1}^{n-1}\mu_j\left[c_j c_{j+1}^* - c_{j+1}^* c_j + c_{j+1}c_j^* - c_j^* c_{j+1} + \gamma_j(c_j c_{j+1} - c_{j+1}c_j + c_{j+1}^* c_j^* - c_j^* c_{j+1}^*)\right] \\ &\quad -\sum_{j=1}^{n}\nu_j(c_j^* c_j - c_j c_j^*) \\ &= \mathcal{C}^*\tilde{M}\mathcal{C}. \end{aligned} \tag{53}$$

where we have also used the anti-commutation properties (18). In the last line we have set

$$\mathcal{C} = (c_1, \ldots, c_n, c_1^*, \ldots, c_n^*)^t, \tag{54}$$

a "column vector", and interpret

$$\mathcal{C}^* = (c_1^*, \ldots, c_n^*, c_1, \ldots, c_n) \tag{55}$$

as a "row vector". We use the block matrix

$$\tilde{M} = \begin{pmatrix} A & B \\ -B & -A \end{pmatrix}, \tag{56}$$

where $A = M$ from (22) and

$$B = \begin{pmatrix} 0 & \gamma_1\mu_1 & & & \\ -\gamma_1\mu_1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & \gamma_{n-1}\mu_{n-1} \\ & & & -\gamma_{n-1}\mu_{n-1} & 0 \end{pmatrix}. \tag{57}$$

Note here that $A^* = A^t = A$ and $B^* = B^t = -B$, and thus $\tilde{M}^* = \tilde{M}^t = \tilde{M}$. Block matrices of the form (56) have many interesting properties (and their appearance in physics is not restricted to the theory of quantum spin chains). One of these properties is that

$$\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\tilde{M}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} = -\tilde{M}, \tag{58}$$

i.e. $\tilde{M}$ is unitarily equivalent to $-\tilde{M}$ and, in particular, $\sigma(\tilde{M}) = -\sigma(\tilde{M})$.

Let $S = A + B$ and $0 \leq \tilde{\lambda}_1 \leq \tilde{\lambda}_2 \leq \ldots \leq \tilde{\lambda}_n$ be the singular values of S, i.e. the eigenvalues of $(S^*S)^{1/2}$, counted with multiplicity. Let $\tilde{\Lambda} := \mathrm{diag}(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_n)$. The singular value decomposition of S gives orthogonal matrices U and V such that

$$USV^t = U(A+B)V^t = \tilde{\Lambda}. \tag{59}$$

This implies

$$\tilde{\Lambda} = \tilde{\Lambda}^t = VS^tU^t = V(A-B)U^t. \tag{60}$$

Let

$$W := \frac{1}{2}\begin{pmatrix} V+U & V-U \\ V-U & V+U \end{pmatrix}. \tag{61}$$

Then W is an orthogonal $2n \times 2n$-matrix. This can be checked by directly verifying that $WW^t = I$, or, alternatively, by noting that

$$SWS^{-1} = \begin{pmatrix} V & 0 \\ 0 & U \end{pmatrix}, \tag{62}$$

with the orthogonal matrix

$$S := \frac{1}{\sqrt{2}}\begin{pmatrix} I & I \\ -I & I \end{pmatrix}. \tag{63}$$

A calculation shows that W diagonalizes $\tilde{M}$ via

$$W\begin{pmatrix} A & B \\ -B & -A \end{pmatrix}W^t = \begin{pmatrix} \tilde{\Lambda} & 0 \\ 0 & -\tilde{\Lambda} \end{pmatrix}. \tag{64}$$

This leads to another interesting property of block matrices of the form (56), namely that

$$S = A + B \text{ is invertible} \iff \text{All } \tilde{\lambda}_j \neq 0 \iff \tilde{M} = \begin{pmatrix} A & B \\ -B & -A \end{pmatrix} \text{ is invertible.} \tag{65}$$

It is easily checked that

$$\mathcal{B} := W\mathcal{C} \tag{66}$$

is of the form

$$\mathcal{B} = (b_1, \ldots, b_n, b_1^*, \ldots, b_n^*)^t. \tag{67}$$

That the c_j, $j = 1, \ldots, n$, satisfy CAR is equivalent to

$$\mathcal{C}\mathcal{C}^* + J(\mathcal{C}\mathcal{C}^*)^t J = I_{2n}, \tag{68}$$

the $2n \times 2n$-identity matrix (where the 1's and 0's which appear are understood as the operators I and 0 on $\mathcal{H}$). Here

$$J := \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix} = J^t.$$

Note that $JW = WJ$ and $JW^t = W^tJ$. By (66),

$$\begin{aligned} \mathcal{B}\mathcal{B}^* + J(\mathcal{B}\mathcal{B}^*)^tJ &= W\mathcal{C}\mathcal{C}^*W^t + J(W\mathcal{C}\mathcal{C}^*W^t)^tJ \\ &= W(\mathcal{C}\mathcal{C}^* + J(\mathcal{C}\mathcal{C}^*)^tJ)W^t \\ &= WI_{2n}W^t \\ &= I_{2n}. \end{aligned} \tag{69}$$

Thus the b_j, $j = 1, \ldots, n$, satisfy CAR as well. Essentially, what we have shown here is the fact that (66) is a *Bogoliubov transformation*.

This Bogoliubov transformation diagonalizes H_γ: By (53) and (64) we have

$$\begin{aligned} H_\gamma &= \mathcal{C}^*\tilde{M}\mathcal{C} = \mathcal{B}^*W\tilde{M}W^t\mathcal{B} = \mathcal{B}^*\begin{pmatrix} \tilde{\Lambda} & 0 \\ 0 & -\tilde{\Lambda} \end{pmatrix}\mathcal{B} \\ &= \sum_{j=1}^n \tilde{\lambda}_j(b_j^*b_j - b_jb_j^*) \\ &= 2\sum_{j=1}^n \tilde{\lambda}_j b_j^*b_j - \tilde{E}_0 I, \end{aligned} \tag{70}$$

where $\tilde{E}_0 = \sum_{j=1}^n \tilde{\lambda}_j$. Thus H_γ has been written in the form of a free fermion system.

As in Section 3 we can argue that the intersections of the kernels of the b_j is one-dimensional, i.e. that they contain an essentially unique vaccum vector $\tilde{\Omega}$ from which an ONB of eigenvectors of H_γ is found as

$$\psi_\alpha = (b_1^*)^{\alpha_1} \ldots (b_n^*)^{\alpha_n}\tilde{\Omega}, \quad \alpha \in \{0,1\}^n. \tag{71}$$

The corresponding eigenvalues are $2\sum_{j:\alpha_j=1} \tilde{\lambda}_j - \tilde{E}_0$, so that one gets

$$\sigma(H_\gamma) = \left\{ 2\sum_{j:\alpha_j=1} \tilde{\lambda}_j - \tilde{E}_0 : \ \alpha \in \{0,1\}^n \right\}. \tag{72}$$

All $\tilde{\lambda}_j$ are non-negative, which leads to expressions for the ground state and ground state energy which, at least formally, look simpler than what was obtained for the isotropic case in Section 3 above: The ground state energy of H_γ is

$$E_0 = -\tilde{E}_0 = -\mathrm{tr}(S^*S)^{1/2}. \tag{73}$$

If $S = A + B$ (or, equivalently by (65), $\tilde{M}$) is invertible, then the vacuum vector $\tilde{\Omega}$ is the unique ground state of H_γ.

For theoretical arguments it is a useful property that the ground state coincides with the vacuum (which was not the case in Section 3 above). From this point of view, the diagonalization procedure described in the current section is superior to the one for the isotropic case described earlier. However, in concrete examples there is also a price to pay, which comes in the form of having to find a the singular value decomposition (59) of the non-self-adjoint matrix $A + B$ instead of just the eigenvalues and eigenvectors of the self-adjoint A.

Remark: Of course, the isotropic XY-chain should arise as a special case of the anisotropic XY-chain. Indeed, the Hamiltonians in (1) and (51) coincide in the case where all $\gamma_j = 0$, which is expressed by the fact that M from (22) reappears as A in (56), while the off-diagonal blocks $\pm B$ in (56) vanish in the isotropic case. The reason that (70) takes a form different from (26) is that the λ_j and $\tilde{\lambda}_j$ are chosen differently. In (26) the λ_j are chosen as eigenvalues of M, while when using (70) in the isotropic case the $\tilde{\lambda}_j$ are eigenvalues of $|A| = |M|$. Essentially, treating the isotropic xy-chain as a special case of the methods introduced for the anisotropic xy-chain amounts to block-diagonalizing $\begin{pmatrix} |M| & 0 \\ 0 & -|M| \end{pmatrix}$ instead of $\begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}$, which are, of course, unitarily equivalent.

We end this section by describing how the arguments on dynamics from Section 4 can be extended to the anisotropic case. For the Heisenberg dynamics of b_k and b_k^* one finds as before

$$\tau_t(b_k) = e^{-2it\lambda_k} b_k \quad \tau_t(b_k^*) = e^{2it\lambda_k} b_k^*, \tag{74}$$

i.e.

$$\tau_t(\mathcal{B}) = \begin{pmatrix} e^{-2it\Lambda} & 0 \\ 0 & e^{2it\Lambda} \end{pmatrix} \mathcal{B}. \tag{75}$$

Furthermore,

$$\begin{aligned} \tau_t(\mathcal{C}) &= e^{itH_\gamma} \mathcal{C} e^{-itH_\gamma} \\ &= e^{itH_\gamma} W^t \mathcal{B} e^{-itH_\gamma} \\ &= W^t e^{itH_\gamma} \mathcal{B} e^{-itH_\gamma} \\ &= W^t \begin{pmatrix} e^{-2it\Lambda} & 0 \\ 0 & e^{2it\Lambda} \end{pmatrix} \mathcal{B} \\ &= W^t \begin{pmatrix} e^{-2it\Lambda} & 0 \\ 0 & e^{2it\Lambda} \end{pmatrix} W\mathcal{C} \\ &= e^{-2it\tilde{M}} \mathcal{C}, \end{aligned} \tag{76}$$

or, writing out the first n components,

$$\tau_t(c_j) = \sum_{\ell=1}^{n} v_{j,\ell}(2t) c_\ell + \sum_{\ell=1}^{n} v_{j,n+\ell}(2t) c_\ell^*, \quad j = 1, \ldots, n, \tag{77}$$

where

$$v_{j,\ell}(t) := \left(e^{-i\tilde{M}t} \right)_{j,\ell}. \tag{78}$$

6 Majorana representation of the XY chain

The specific structure of $\tilde{M}$ and (77) suggests the change of basis $(e_1, e_2, \ldots, e_{2n})$ to $(e_1, e_{n+1}, e_2, e_{n+2}, \ldots, e_n, e_{2n})$ in representing $\tilde{M}$, as this better reflects locality relations between the observables c_j, c_j^* as well as a_j, a_j^*. This change of basis is implemented by the $2n \times 2n$-permutation matrix P defined by

$$Pe_{2j-1} = e_j, \quad Pe_{2j} = e_{n+j}, \quad j = 1, \ldots, n. \tag{79}$$

In the new basis we have

$$\tilde{\mathcal{A}} := P^t \tilde{M} P = \begin{pmatrix} -\nu_1\sigma^Z & \mu_1 S(\gamma_1) & & & \\ \mu_1 S(\gamma_1)^t & -\nu_2\sigma^Z & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & \mu_{n-1}S(\gamma_{n-1}) \\ & & & \mu_{n-1}S(\gamma_{n-1})^t & -\nu_n\sigma^Z \end{pmatrix}. \tag{80}$$

Here σ^Z is the Pauli matrix from above and

$$S(\gamma) := \begin{pmatrix} 1 & \gamma \\ -\gamma & -1 \end{pmatrix}. \tag{81}$$

Thus $\tilde{\mathcal{A}}$ takes the form of a Jacobi matrix with 2×2-matrix-valued entries

$$\tilde{\mathcal{A}}_{jk} = \begin{cases} -\nu_j\sigma^Z & \text{if } j = k, \\ \mu_j S(\gamma_j) & \text{if } k = j+1, \\ \mu_{j-1}S(\gamma_{j-1})^t & \text{if } k = j-1, \end{cases} \tag{82}$$

and zero for all other j, k. In terms of $\tilde{\mathcal{A}}$ the Hamiltonian H_γ can be rewritten as

$$H_\gamma = \sum_{j,k} (c_j^*, c_j) \tilde{\mathcal{A}}_{jk} \begin{pmatrix} c_k \\ c_k^* \end{pmatrix}. \tag{83}$$

From (80) one can proceed to another change of basis which leads us into a discussion of the *Majorana representation* of the XY chain.

For this we start by introducing a set of $2n$ so-called *Majorana operators* as

$$\tilde{c}_{2j-1} := \sigma_1^Z \dots \sigma_{j-1}^Z \sigma_j^X / \sqrt{2}, \tag{84}$$

$$\tilde{c}_{2j} := \sigma_1^Z \dots \sigma_{j-1}^Z \sigma_j^Y / \sqrt{2}, \tag{85}$$

$j = 1, \dots, n$. The Majorana operators are self-adjoint and closely related to the Fermionic annihilation and creation operators c_j and c_j^* defined in (15) and (16),

$$c_j = \frac{1}{\sqrt{2}}(\tilde{c}_{2j-1} - i\tilde{c}_{2j}), \quad c_j^* = \frac{1}{\sqrt{2}}(\tilde{c}_{2j-1} + i\tilde{c}_{2j}), \tag{86}$$

or

$$\tilde{c}_{2j-1} = \frac{1}{\sqrt{2}}(c_j + c_j^*), \quad \tilde{c}_{2j} = \frac{i}{\sqrt{2}}(c_j - c_j^*). \tag{87}$$

With

$$R := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \tag{88}$$

we can rewrite (86) and (87) as

$$\begin{pmatrix} c_j \\ c_j^* \end{pmatrix} = R \begin{pmatrix} \tilde{c}_{2j-1} \\ \tilde{c}_j \end{pmatrix}, \quad (c_j^*, c_j) = (\tilde{c}_{2j-1}, \tilde{c}_{2j}) R^*. \tag{89}$$

We note that the factor $1/\sqrt{2}$ in the definition of the Majorana operators is usually omitted in the physics literature. It is included here because it makes R unitary.

Using this together with (82) and (83) we can represent the Hamiltonian of the anisotropic XY chain in terms of the Majorana operators as

$$H_\gamma = \frac{i}{2}(\tilde{c}_1, \ldots, \tilde{c}_{2n})\, \mathcal{A} \begin{pmatrix} \tilde{c}_1 \\ \vdots \\ \tilde{c}_{2n} \end{pmatrix}, \tag{90}$$

where $\mathcal{A}$ is the $2n \times 2n$ real skew-symmetric matrix

$$\mathcal{A} = \begin{pmatrix} A_1 & B_1 & & \\ -B_1^t & A_2 & \ddots & \\ & \ddots & \ddots & B_{n-1} \\ & & -B_{n-1}^t & A_n \end{pmatrix} \tag{91}$$

with 2×2-matrix-valued entries

$$A_j = \begin{pmatrix} 0 & \nu_j \\ -\nu_j & 0 \end{pmatrix}, \quad B_j = \mu_j \begin{pmatrix} 0 & -(1-\gamma_j) \\ 1+\gamma_j & 0 \end{pmatrix}. \tag{92}$$

Note that the Majorana representation takes a particularly simple form for the case of the Ising model, where all $\gamma_j = 1$. In this case the matrix $\mathcal{A}$ is tri-diagonal with zero entries in the main diagonal.

Of course, $\mathcal{A}$ and $\tilde{\mathcal{A}}$ are closely related to $\tilde{M}$. From (80) and (64) we find

$$\tilde{\mathcal{A}} = \tilde{W}^t \begin{pmatrix} \tilde{\lambda}_1 \sigma^Z & & \\ & \ddots & \\ & & \tilde{\lambda}_n \sigma^Z \end{pmatrix} \tilde{W}, \tag{93}$$

where $\tilde{W} := P^t W P$. This in turn yields

$$\mathcal{A} = -2i \begin{pmatrix} R^* & & \\ & \ddots & \\ & & R^* \end{pmatrix} \tilde{W}^t \begin{pmatrix} \tilde{\lambda}_1 \sigma^Z & & \\ & \ddots & \\ & & \tilde{\lambda}_n \sigma^Z \end{pmatrix} \tilde{W} \begin{pmatrix} R & & \\ & \ddots & \\ & & R \end{pmatrix}. \tag{94}$$

In particular, the eigenvalues of $\mathcal{A}$ are $\pm 2i\tilde{\lambda}_j$, $j = 1, \ldots, n$, where $\pm\tilde{\lambda}_j$, $j = 1, \ldots, n$, are the pairs of eigenvalues of $\tilde{M}$.

7 Entanglement entropy of the ground state

Our goal in this section is to derive an explicit formula for the entanglement of the ground state of the anisotropic XY chain in random exterior field. This formula was first presented in [9], with a more detailed derivation given in [4]. The presentation provided here differs in two aspects from these works: Instead of using Majorana operators to express the results, as

done in [9] and [4], we work exclusively in the language of Fermionic creation and annihilation operators. The facts presented in Section 6 allow to easily switch between these two languages (working with Fermionic operators essentially means that we prefer self-adjoint operators over skew-adjoint operators). Also, while [9] and [4] present their results for the constant coefficient case, we will continue to use the general variable coefficient model (49).

We will use the notation and results of Section 5, but write λ_j and Ω instead of $\tilde{\lambda}_j$ and $\tilde{\Omega}$ for the singular values of S and normalized ground state of H_γ. The latter is, up to a phase, unique, as we will assume that all λ_j are non-zero. We will write $\rho = |\Omega\rangle\langle\Omega|$ for the ground state projection and $\langle\cdot\rangle_\rho := \langle\Omega|\cdot|\Omega\rangle$ for ground state expectations.

We have $b_k\Omega = 0$ for all k and thus

$$\langle b_k\rangle_\rho = \langle b_k^*\rangle_\rho = 0 \quad \text{for all } k. \tag{95}$$

Furthermore, for all k and j,

$$\langle b_k b_j\rangle_\rho = \langle b_k^* b_j^*\rangle_\rho = \langle b_k^* b_j\rangle_\rho = 0, \quad \langle b_k b_j^*\rangle_\rho = \delta_{kj}, \tag{96}$$

where the latter uses the CAR. This can be conveniently summarized in form of the correlation matrix of the Fermionic operators $\mathcal{B} = (b_1, \ldots, b_n, b_1^*, \ldots, b_n^*)$ with respect to the ground state ρ:

$$\Gamma_\rho^{\mathcal{B}} := \langle \mathcal{B}\mathcal{B}^*\rangle_\rho = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}. \tag{97}$$

Here $\langle \mathcal{B}\mathcal{B}^*\rangle_\rho$ is to be understood in the sense of taking expectations of each of the operator-valued entries of $\mathcal{B}\mathcal{B}^*$.

We note here that the state ρ is uniquely determined by the correlation matrix $\Gamma_\rho^{\mathcal{B}}$ and (95). This is due to Wick's Rule which allows to determine $\langle A\rangle_\rho$ from (95) and (97) for each A which is a product of finitely many of the b_j and b_j^*. From the properties of free Fermion systems listed in Section 3 one sees that this includes the rank-one orthogonal projections onto each one of the basis states ψ_α from (27), and thus spans all of $B(\mathcal{H})$. Thus $\langle A\rangle_\rho$ is determined for all $A \in B(\mathcal{H})$, which in turn determines ρ (using that $A \mapsto \langle A\rangle_\rho = \mathrm{tr}(\rho A)$ is a non-trivial linear functional on $B(\mathcal{H})$).

Next we can find the correlation matrix of $\mathcal{C} = W^t\mathcal{B}$ with respect to ρ. It is easily checked that

$$\Gamma_\rho^{\mathcal{C}} := \langle \mathcal{C}\mathcal{C}^*\rangle_\rho = \langle W^t\mathcal{B}\mathcal{B}^*W\rangle_\rho = W^t\langle \mathcal{B}\mathcal{B}^*\rangle_\rho W = W^t\Gamma_\rho^{\mathcal{B}}W. \tag{98}$$

Now a calculation using (61), (97) and orthogonality of V and U gives

$$\Gamma_\rho^{\mathcal{C}} = \frac{1}{2}I + \begin{pmatrix} X & Y \\ -Y & -X \end{pmatrix}, \tag{99}$$

where

$$X = \frac{1}{4}(U^tV + V^tU), \quad Y = \frac{1}{4}(U^tV - V^tU). \tag{100}$$

As $\Gamma_\rho^{\mathcal{C}}$ is orthogonally equivalent to $\Gamma_\rho^{\mathcal{B}}$ it is self-adjoint and has eigenvalues 0 and 1, both with n-dimensional eigenspaces. This means that the self-adjoint block matrix $\begin{pmatrix} X & Y \\ -Y & -X \end{pmatrix}$ has eigenvalues $1/2$ and $-1/2$ with n-dimensional eigenspaces.

Our goal is to find a formula for the entanglement entropy $\mathcal{E}_\ell(\Omega)$ of the ground state Ω with respect to the decomposition of $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, where

$$\mathcal{H}_1 = \bigotimes_{j=1}^{\ell} \mathbb{C}^2, \quad \mathcal{H}_2 = \bigotimes_{j=\ell+1}^{n} \mathbb{C}^2, \tag{101}$$

corresponding to a decomposition of the spin chain into the ℓ sites at the left end of the chain and the remaining sites. Consider the reduction of the ground state $\rho_\ell := \mathrm{tr}_{\mathcal{H}_2}\rho$ to $\mathcal{H}_1$. Then the entanglement entropy is given by

$$\mathcal{E}_\ell(\Omega) := -\mathrm{tr}(\rho_\ell \log \rho_\ell), \tag{102}$$

the von Neumann entropy $S(\rho_\ell)$ of the reduced state ρ_ℓ.

The expectation of $A \in B(\mathcal{H}_1)$ with respect to the mixed state ρ_ℓ is, as usual, defined as

$$\langle A \rangle_{\rho_\ell} = \mathrm{tr}(\rho_\ell A). \tag{103}$$

Let $\mathcal{C}_\ell := (c_1, \ldots, c_\ell, c_1^*, \ldots, c_\ell^*)^t$, where we consider c_j, c_j^*, $j = 1, \ldots, \ell$, as operators on $\mathcal{H}_1$, which is possible due to their definition via the 'left-anchored' Jordan-Wigner transform (15) (thus we abuse notation by identifying $c_j = c_j \otimes I$ for these operators). $\mathcal{C}_\ell$ is a maximal set of Fermionic operators for $\mathcal{H}_1$. We define the $2\ell \times 2\ell$ correlation matrix of $\mathcal{C}_\ell$ with respect to ρ_ℓ as

$$\Gamma_{\rho_\ell}^{\mathcal{C}_\ell} = \langle \mathcal{C}_\ell \mathcal{C}_\ell^* \rangle_{\rho_\ell}, \tag{104}$$

where expectations are again taken component-wise.

We claim that

$$\Gamma_{\rho_\ell}^{\mathcal{C}_\ell} = \frac{1}{2} I_{2\ell} + \begin{pmatrix} X_\ell & Y_\ell \\ -Y_\ell & -X_\ell \end{pmatrix}, \tag{105}$$

where X_ℓ and Y_ℓ are the upper right $\ell \times \ell$-blocks of X and Y.

This follows from the following general fact: If $A \in B(\mathcal{H}_1)$, $\rho \in B(\mathcal{H}_1 \otimes \mathcal{H}_2)$ and $\rho_1 = \mathrm{tr}_{\mathcal{H}_2}\rho$, then

$$\mathrm{tr}_{\mathcal{H}_2}((A \otimes I)\rho) = A\, \mathrm{tr}_{\mathcal{H}_2}\rho, \tag{106}$$

which is seen by a simple calculation using the definition of the partial trace. Inserting a pure state $\rho = |\Omega\rangle\langle\Omega|$ into (106) one gets

$$\langle A \rangle_{\rho_1} = \mathrm{tr}(A\rho_1) = \mathrm{tr}(A(\mathrm{tr}_{\mathcal{H}_2}\rho)) = \mathrm{tr}(\mathrm{tr}_{\mathcal{H}_2}((A \otimes I)\rho)) = \mathrm{tr}((A \otimes I)\rho) = \langle \Omega | A \otimes I | \Omega \rangle. \tag{107}$$

From this we see that (105) follows from (99), as all entries of $\mathcal{C}_\ell \mathcal{C}_\ell^*$ are in $B(\mathcal{H}_1)$.

As X_ℓ is self-adjoint and Y_ℓ skew-adjoint, the matrix $\begin{pmatrix} X_\ell & Y_\ell \\ -Y_\ell & -X_\ell \end{pmatrix}$ can be diagonalized in the same way as the matrix block matrix $\tilde{M}$ in (56). We have $S_\ell := X_\ell + Y_\ell = \frac{1}{2}(U^t V)_\ell$ and

$$0 \leq S_\ell^* S_\ell = \frac{1}{4}((U^t V)_\ell)^t (U^t V)_\ell \leq \frac{1}{4} I. \tag{108}$$

The latter uses a simple general fact for the restriction of orthogonal matrices (in this case $U^t V$) to subspaces.

It thus follows that $(S_\ell^* S_\ell)^{1/2}$ has eigenvalues $0 \leq \mu_j \leq 1/2$, $j = 1, \ldots, \ell$, and therefore S_ℓ the singular-value decomposition

$$U_\ell S_\ell V_\ell^t = \Lambda_\ell := \mathrm{diag}(\mu_j, j = 1, \ldots, \ell), \tag{109}$$

where U_ℓ and V_ℓ are $2\ell \times 2\ell$ orthogonal matrices.

By the arguments from Section 5 we get

$$W_\ell \begin{pmatrix} X_\ell & Y_\ell \\ -Y_\ell & -X_\ell \end{pmatrix} W_\ell^t = \begin{pmatrix} \Lambda_\ell & 0 \\ 0 & -\Lambda_\ell \end{pmatrix}, \tag{110}$$

with

$$W_\ell := \frac{1}{2} \begin{pmatrix} V_\ell + U_\ell & V_\ell - U_\ell \\ V_\ell - U_\ell & V_\ell + U_\ell \end{pmatrix}. \tag{111}$$

In particular, we conclude that the self-adjoint correlation matrix (105) satisfies

$$0 \leq \Gamma_{\rho_\ell}^{\mathcal{C}_\ell} \leq I \tag{112}$$

and has eigenvalues

$$\frac{1}{2} \pm \mu_j, \quad j = 1, \ldots, \ell. \tag{113}$$

The central claim of this section is that these eigenvalues allow to find the entanglement entropy of Ω with respect to $\mathcal{H}_1$:

Proposition 7.1. *For all $1 \leq \ell \leq n$ it holds that*

$$\mathcal{E}_\ell(\Omega) = -\sum_{j=1}^{\ell} \left(\left(\frac{1}{2} + \mu_j \right) \log \left(\frac{1}{2} + \mu_j \right) + \left(\frac{1}{2} - \mu_j \right) \log \left(\frac{1}{2} - \mu_j \right) \right). \tag{114}$$

I don't yet fully understand how to make the argument for this from [4] and [9] rigorous. However, in the following are a couple of reasons which might lead to a proof.

(i) To diagonalize $\Gamma_{\rho_\ell}^{\mathcal{C}_\ell}$ we introduce a new set of operators satisfying the CAR as

$$\tilde{\mathcal{B}}_\ell := W_\ell \mathcal{C}_\ell = (\tilde{b}_1, \ldots, \tilde{b}_\ell, \tilde{b}_1^*, \ldots, \tilde{b}_\ell^*)^t. \tag{115}$$

By an argument similar to (98) (which also works for mixed states) we find for the new correlation matrix $\Gamma_{\rho_\ell}^{\tilde{\mathcal{B}}_\ell} = \langle \tilde{\mathcal{B}}_\ell \tilde{\mathcal{B}}_\ell^* \rangle_{\rho_\ell}$ that

$$\Gamma_{\rho_\ell}^{\tilde{\mathcal{B}}_\ell} = W_\ell \Gamma_{\rho_\ell}^{\mathcal{C}_\ell} W_\ell^t = \frac{1}{2} I_{2\ell} + W_\ell \begin{pmatrix} X_\ell & Y_\ell \\ -Y_\ell & -X_\ell \end{pmatrix} W_\ell^t = \begin{pmatrix} \frac{1}{2} I_\ell + \Lambda_\ell & 0 \\ 0 & \frac{1}{2} I_\ell - \Lambda_\ell \end{pmatrix}. \tag{116}$$

(ii) In [4] it is argued that (116) means that ρ_ℓ is a product state. What we will show here, instead, is that there is a product state in $\mathcal{H}_1$, which leads to the same correlation matrix as (116) for the representation $\mathcal{C}_\ell$ of the CAR.

For this, choose numbers $0 \le \xi_j \le 1$, $j = 1, \ldots, \ell$, and consider

$$\tilde{\rho}_\ell = \bigotimes_{j=1}^{\ell} \begin{pmatrix} \xi_j & 0 \\ 0 & 1-\xi_j \end{pmatrix}. \tag{117}$$

This is a product state with von Neumann entropy

$$S(\tilde{\rho}_\ell) = \sum_{j=1}^{\ell} S\left(\begin{pmatrix} \xi_j & 0 \\ 0 & 1-\xi_j \end{pmatrix} \right) = -\sum_{j=1}^{\ell} (\xi_j \log \xi_j + (1-\xi_j)\log(1-\xi_j)). \tag{118}$$

On the other hand, with c_j defined as above via the Jordan-Wigner transform, it is not hard (although somewhat tedious) to find the correlation matrix $\Gamma^{\mathcal{C}_\ell}_{\tilde{\rho}_\ell}$ (all calculations are straightforward, because $\tilde{\rho}_\ell$ as well as all c_j and c_j^* are product operators and thus all the needed traces are products of traces of two-by-two matrices). One gets

$$\Gamma^{\mathcal{C}_\ell}_{\tilde{\rho}_\ell} = \mathrm{diag}(1-\xi_1, \ldots, 1-\xi_\ell, \xi_1, \ldots, \xi_\ell). \tag{119}$$

Thus, with the choice $\xi_j = \frac{1}{2} - \mu_j$, $j = 1, \ldots, \ell$, we find that the diagonal correlation matrixes $\Gamma^{\tilde{\mathcal{B}}_\ell}_{\rho_\ell}$ and $\Gamma^{\mathcal{C}_\ell}_{\tilde{\rho}_\ell}$ are identical.

The question which remains is if this shows that the two mixed states ρ_ℓ and $\tilde{\rho}_\ell$ are unitarily equivalent (or, in other words, have the same eigenvalues), meaning that they also have the same von Neumann entropies. Then (114) would follow from (118). My best guess right now is that this is well known and one should be able to find a reference for it.

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